

Field	Details
Date	19 July 2026
Team ID	SB-AIML-2026-0042
Project Name	Emotion Detector - AI-Driven Emotion Detection & Personalized Learning Support P
Maximum Marks	3 Marks

Project Timeline (Gantt View)

Phase	Module	Start Date	End Date	Duration	Status
1	Environment Setup	01 June 2026	03 June 2026	3 days	Done
2	Data Acquisition &	04 June 2026	08 June 2026	5 days	Done
3	BiLSTM Model	09 June 2026	14 June 2026	6 days	Done
4	BERT Model	15 June 2026	21 June 2026	7 days	Done
5	Detection Engine	22 June 2026	25 June 2026	4 days	Done
6	Gemini Integration	26 June 2026	28 June 2026	3 days	Done
7	Streamlit Web	29 June 2026	05 July 2026	7 days	Done
8	Analytics & Logging	06 July 2026	08 July 2026	3 days	Done
9	Testing	09 July 2026	11 July 2026	3 days	Done
10	Deployment	12 July 2026	12 July 2026	1 day	Done
11	Documentation &	13 July 2026	19 July 2026	7 days	Done
Total Duration: ~49 days		Submission Prep			

Milestone	Target Date	Deliverable	Status
M1 - Data Ready	08 June 2026	Cleaned dataset, train/val/test	
M2 - Models Trained	21 June 2026	BiLSTM + BERT weights saved	
M3 - Detection Engine Live	25 June 2026	Unified detect() interface	
M4 - Guidance Working	28 June 2026	Gemini returns 5-section	
M5 - App Running Locally	05 July 2026	Full Streamlit app runs	
M6 - Deployed to Cloud	12 July 2026	Public URL live on Streamlit	
M7 - Documentation Done	19 July 2026	All 8 phases documented for submission	

Resource Allocation

Resource	Allocation	Tools Used
Developer	100% (Solo Project - Harsh)	Python, VS Code, GitHub
Compute (Training)	Local GPU / CPU	TensorFlow, PyTorch
API Credits	Google Gemini Free Tier	1500 req/day
Deployment	Streamlit Community Cloud	Free
Dataset	HuggingFace Hub	dair-ai/emotion (16K samples)

Risk Register

S.No	Risk	Likelihood	Impact	Mitigation
1	Gemini API rate limit hit	Medium	High	Implement retry with exponential backoff; add

				fallback message
2	BiLSTM underfitting on small dataset	Medium	Medium	Class weighting, early stopping, data augmentation
3	BERT OOM on low-RAM machine	Low	High	Use batch size 8, gradient accumulation, CPU fallback
4	Streamlit Cloud cold start delay	High	Low	Inform users via loading spinner; use session state caching
5	Static PDF templates not accepted	Medium	High	Generate PDFs from Markdown using VS Code/Pandoc

Activity	Frequency	Channel
Progress tracking	Daily	GitHub commit log
Milestone review	Weekly	Self-review checklist
Issue tracking	As needed	GitHub Issues
Submission check	Final	Course portal